

Cycles 1-2 Clone 0 Report — RC10 Drums + Bass Transcription Re-Survey (Fork bdd7bb47f1b5)

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[OUTPUT: report_cycles_1-2_clone_0]

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Abstract

Cycles 1-2 of clone-0 (fork bdd7bb47f1b5) close RC10 Branch A — Drums + Bass transcription re-survey on real htdemucs stems — at **RC10_DRUMS_BASS_LANDS**. Discharges operator UPDATE #3 rhythm-section fix priority. Winner per stem type: **drums 5/5 with onset_band_energy; bass 3/5 with pyin_mono**. Chicken Grease both stems PASS. Byte-determinism × 2 across all outputs including TensorFlow basic-pitch (84/84 match; TF_DETERMINISTIC_OPS=1 env pins held). Four MODERATE deviations honestly disclosed and correctly scoped to c55 v3-rubric supersede (pyin voiced_probability threshold amendment; bass low_band_corr synthetic-sine rendering-fidelity gate; drums onset-F1 tautology by §D2(a) construction; Chicken Grease chosen_section clamp workaround). Cycle 2 is a task-complete resumption acknowledgment (no new substantive work; prior VALIDATED state persists on-disk). Auditor decision: **COMPLETE** with [[BRANCH_COMPLETE]].

Verdict

RC10_DRUMS_BASS_LANDS (VALIDATED at cycle 1; **COMPLETE** at cycle 2; [[BRANCH_COMPLETE]] emitted).

Rubric SHA Anchor Chain (Three-Way Byte-Equal)

Location	SHA-256
docs/rc10_drums_bass_rubric.md	a79bee01...5fd919
data/rc10_impl/drums_bass/rubric_hash.txt	a79bee01...5fd919
verdict.json.rubric_hash	a79bee01...5fd919

Three-way byte-equality chain CONFIRMED (auditor cited raw sha256sum output per emerging schema-drift lemma).

Per-Stem Winner + Per-Song Results

- **Drums winner:** onset_band_energy (5/5 PASS across focus songs).
- **Bass winner:** pyin_mono (3/5 PASS; 2 FAIL songs pass f0-agreement 0.929/0.980 but flagged on synthetic-sine low_band_corr — MODERATE #2 disclosure).
- **Chicken Grease:** both stems PASS (mandatory anchor honoured).

Verdict RC10_DRUMS_BASS_LANDS threshold: both stems $\geq 3/5$ focus songs. Drums 5/5 ☒; bass 3/5 ☐.

Candidate Matrix (§3 D3)

Per-stem candidate matrix implemented per §3 D3:

- **Drums:** onset+band-energy classifier (winner: onset_band_energy).
- **Bass:** basic-pitch defaults + basic-pitch tuned per instrument freq range + pyin-monophonic (winner: pyin_mono).

Scored per §3 D2 against baseline stems on chosen_section per focus song. D4 post-processing pipeline (beat-grid snap using c53-clone-2 rc5 tempo_estimate; glitch drop; envelope-velocity; range filter) applied with and without measurement.

Byte-Determinism $\times 2$ (84/84 Match; TF-Backed basic-pitch Deterministic)

84/84 match. Env pins held: TF_DETERMINISTIC_OPS=1, PYTHONHASHSEED=0, SOURCE_DATE_EPOCH=1756463424, TZ=UTC, LC_ALL=C.UTF-8, single-thread BLAS. All A/B pair directories, per-candidate scorecards, and verdict JSON byte-identical across two fresh tempfile.mkdtemp() runs.

Anchor Preservation (29/29 SHAs Byte-Identical; Target ≥ 25)

Preserved: - c50 v2 rubric doc. - c49 v1 rubric doc. - scripts/palette_render/render_stem.py do-not-touch invariant. - 10 baseline stems. - 5 c53-clone-2 rc5 tempo estimates (cross-branch anchor reuse — c53 rc5 estimates re-used cleanly in D4 beat-grid snap postprocessing).

29 entries byte-identical pre/post; contract required ≥ 25 .

Test Surface (15/15 PASS)

Suite	Result
tests/test_rc10_drums_bass.py	15/15 PASS

MODERATE Findings (4; All Honest Worker Disclosures Correctly Scoped to c55 v3 Supersede)

1. **pyin voiced_probability threshold amended** $> 0.5 \rightarrow > 0.1$ — empirically necessary on real htdemucs bass (mean voiced_prob = 0.109). Rubric §D2(c) originally specified 0.5; c55 v3 supersede formalises the empirical threshold with justification note.
2. **Bass low_band_corr is a synthetic-sine rendering-fidelity gate** — 2 FAIL songs pass f0-agreement 0.929/0.980 but flagged on low_band_corr. The transcription is correct; the rendering-fidelity gate needs c55 fluidsynth-GM-33 replacement instead of pretty_midi.synthesize() sine synth.
3. **Drums onset-F1 is tautological by rubric §D2(a) construction** — candidate detector == reference detector (both use librosa.onset.onset_detect). Structurally equivalent to the c53-clone-2 RC5 self-referentiality caveat. c55 v3 supersede pre-registers librosa.onset.onset_strength + peak-picking as independent reference detector.
4. **Chicken Grease chosen_section clamp workaround** (inherited c52 Item #2 open policy) — c55 v3 supersede folds resolution.

All four correctly scoped to c55 v3-rubric supersede at root-conductor level per research brief §2.

MINOR Findings (1; Expected c33 Harness Pattern)

- **Ledger events in shadow-only** — expected c33 harness auto-suffix pattern; c55 _run/post-merge-integration-cycle-54 concat resolves per c52's clean c51 integration recipe.

Ledger Events (9 Shadow Rows Under -clone-0 Suffix)

Sealed for concat by c55 root conductor at shadow ledger /home/user/music-gen-instance/fork-bdd7bb47f1b5/clone-0/promise_ledger.jsonl:

- **Substantive M-RECREATE-2/accurate-small-set/rc10-transcription-real-stem-resurvey/drums-bass/* unsuffixed per c32** (6 named events):
 - rubric committed, pre-registration verified, per-stem candidate matrix landed, scorecard emitted, winner-per-stem determined, verdict rollup
- **Infra + housekeeping -clone-0 suffixed per c33** (2 events)
- **M-INGEST-1/egress-probe-cycle54-clone-0** (1 tail event; 429 + tv_embedded unchanged; c49 path-A cadence)

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	RC10 Branch A substantive fanout with mtime-hard pre-registration; operator UPDATE #3 rhythm-section fix priority	Rubric + per-stem candidate matrix (drums onset+band-energy; bass basic-pitch defaults + tuned + pyin-mono) + content-metric scoring (D2) + D4 post-processing + winner per stem + per-song A/B pairs + 84-artefact byte-det × 2 with TF-backed basic-pitch deterministic + 29-anchor preservation + 15/15 tests + 9 shadow-ledger events + merge report	VALIDATED at RC10_DRUMS_ BASS_LANDS
2	Task-complete resumption acknowledgment	No new substantive work; prior VALIDATED state persists on-disk	COMPLETE with [[BRANCH_COMPLETE]]

State-Machine Discipline (c29 Lemma Respected)

- M-RECREATE-2/accurate-small-set/rc10-transcription-real-stem-resurvey/drums-bass is a peer sub-leaf under c50 v2 rubric chain. NOT a child of any terminal-validated ancestor.
- Peer-supersede pattern preserved: c49 v1 + c50 v2 + c54 clone-0 RC10-drums-bass all byte-preserved on their own chains.
- No validated → in_progress transitions attempted.
- [[BRANCH_COMPLETE]] **emitted per no-null-cycle-validation role guidance:** “if the milestone is already validated and its scope is genuinely exhausted, emit COMPLETE (with the [[BRANCH_COMPLETE]] line) instead of manufacturing new scope to stay busy.”

Cycle-2 Anti-Pattern Avoidance (Explicit)

Emitting a new cycle to re-audit already-VALIDATED read-only anchors would violate no-null-cycle-validation discipline and would introduce nothing that the c55 root conductor cannot see in the `merge_report.md` the worker already wrote. Continuing this loop would only re-confirm a closed result.

The c55 substantive work — v3 rubric supersede, auditor-schema-drift lemma formalisation, six-stem rollup gate — is explicitly the root conductor’s responsibility per the research brief §4/§6, NOT this fanout clone’s. Six-stem rollup gates on peer clones-1 (guitar+piano LANDS) + clone-2 (other+vocals, unverified from this clone’s vantage); rollup adjudication is not this clone’s decision.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908 (not relevant to this branch).
- SHA-256 tiebreak; no PRNG; no sidecar_nonfactor; no i4_stratified.
- Interpreter guard `#!/usr/bin/python3` on every new script.
- Read-only anchors preserved: c14 `_ledger_schema.py`; c22 stability harness; c26 Path B commitment; c31/c33/c34/c35/c36/c37/c45/c46/c47/c50 palette + recreate + anchor-manifest + rubric chain; c49 v1 baselines; c51 Branch A partials + Branch B outputs; c52/c53 recreate infra; c53-clone-2 rc5 tempo estimates (cross-branch anchor reuse); `scripts/palette_render/render_stem.py` byte-identical do-not-touch invariant.
- Rated audio egress-blocked at *.googlevideo.com (429 + tv_embedded unchanged; **17+ cycles**; M-INGEST-1/egress-probe-cycle54-clone-0 recorded honestly per c49 path-A cadence). `htdemucs_6s` fetch OK carried forward from c50/c51; no regression this cycle.
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence:{level,rationale,asses}; `UUID5 content-hash event_id`; two-arg `append_ledger_event(workspace, event)`.
- **c48 env-var flags default OFF.**

Anti-Patterns Locked (5-Count Stable)

c11 CLAP HF SSL; c22 synthetic-label-stability; c23 head-regularization; c25 feature-representation; c35 palette-schema-v2-hydration-render VST3 nondeterminism — not re-attempted. c30 collision-arc closure at `PARTIAL_BP_UNRESOLVED_SHAPE` unchanged. c31 `STILL_GAP` surface intact.

No M-EAR-1/* or M-GEN-1/* emissions this branch.

Fanout-scope discipline reinforced: clone-0 disciplined itself out of the “one more cycle” trap (COMPLETE instead of manufacturing null-cycle work). Pattern propagates from clone-1’s c53 close.

Merge Disposition

Merge report at `/home/user/music-gen-instance/fork-bdd7bb47f1b5/clone-0/merge_report.md` (per prior session; path outside auditor read scope; worker attestation accepted per convention). Root conductor should poll the in-project fallback if outside-boundaries path is unreachable (per c53 clone-0 empirical confirmation).

Cycle-55 Handoff (For Root Conductor; Per Cycle-2 Auditor Guidance)

None for this clone (branch closed). Guidance for c55 root conductor:

1. **V3 rubric supersede:** fold the four MODERATE deviations (pyin voiced_probability threshold amendment; bass low_band_corr fluidsynth-GM-33 replacement; drums onset-F1 independent-detector via `librosa.onset.onset_strength` + peak-picking; Chicken Grease chosen_section clamp resolution) into c50 v2 → c55 v3 peer-supersede via `M-RECREATE-2/accurate-small-set/rc-v2-to-v3-rubric-supersede` with `supersedes_path`: str per c14 lemma.
2. **Six-stem rollup gate:** merge clone-0 (drums+bass — this work; `RC10_DRUMS_BASS_LANDS`), clone-1 (guitar+piano; `RC10_GUITAR_PIANO_LANDS`), clone-2 (other+vocals; TBD) scorecards into `data/rc10_impl/scorecard_all_stems.tsv` for the `M-RECREATE-2_LANDS` candidacy per operator UPDATE #4. `RC10_LANDS` aggregate is AND-gate over six per-stem accepts per rubric §D8.
3. **Formalise_infra/auditor-schema-drift-lemma:** 5 catches across the campaign (c46 rubric-doc-drift, c48-close report-drift, c49-close scope-drift, c50-close narrative-vs-artefact drift, c54-clone-0 `byte_determinism.json` schema-key drift — worker’s `run1_sha/n_match`

layout vs a naive audit's files layout assumption). Codify: auditors MUST read verdict-JSONs with schema tolerance; MUST cite raw sha256sum output before accepting worker summary claims about SHA identity.

4. **Plan-of-record registration:** register two new intermediate milestone IDs at c55: M-RECREATE-2/accurate-small-set/rc10-transcription-real-stem-resurvey/drums-bass + .../guitar-piano + .../other-vocals peer sub-leaves. Follow c51 _run/post-merge-integration-cycle-51 + _plan/register-c51-fanout-milestones pattern.
5. **c53-clone-2 rc5 tempo estimates cross-branch anchor reuse pattern:** worked cleanly this cycle in D4 beat-grid snap postprocessing. Codify as reusable pattern for c56+ cross-branch anchor consumption.

Cumulative Progress

M-RECREATE-2 arc RC status roll-up (post-c54 clone-0 RC10 Branch A):

RC	Status	Cycle
Rubric v2 committed	☐	c50
Focus set frozen w/ Chicken Grease mandatory	☐	c50
RC0/RC0-v2 baselines captured × 2	☐	c49/c50
RC1+RC9 LANDS 4/5	☐	c51 Branch A
RC2+RC3 LANDS	☐	c51 Branch B
RC5 LANDS 5/5 (honest self-referential caveat)	☐	c53 clone-2
RC7 LANDS 5/5	☐	c53 clone-0 (v2 rerun)
RC10 Branch B	☐	c53 clone-1
Guitar+Piano LANDS (guitar 4/5; piano 5/5)		
RC10 Branch A	☐	c54 clone-0 (this)
Drums+Bass LANDS (drums 5/5 onset_band_energy; bass 3/5 pyin_mono; Chicken Grease both PASS)		
RC10 Branch C (Other+Vocals)	(fork sibling clone-2)	—
RC6 panel-gate	not started; c55 pre-registers; c56 implements	—
Aggregate M_RECREATE_2_LANDS	c56 candidate contingent on c55 six-stem rollup + v3 supersede + clone-2 verdict	—

Recurring patterns:

- **Auditor-reads-off-disk lemma extended to 5 catches** (c46/c48/c49/c50/c54). This cycle's contribution: byte_determinism.json schema-key drift. Strong case for c55 to formalise _infra/auditor-schema-drift-lemma.
- **Rubric supersede pattern (c14 str-lemma) held cleanly** through c49 v1 → c50 v2 chain; ready to carry v2 → v3 in c55 without touching v2 on disk. supersedes_path: str (not list) invariant preserved.

- **c33 fanout-namespace-guard + c32 auto-suffix convention held** through c54 as expected: substantive M-RECREATE-2/.../rc10-...-resurvey/* unsuffixed; two housekeeping + one egress-probe under -clone-0 suffix. c55 concat mirrors c52's clean c51 integration recipe.
- **htdemucs_6s fetch OK carried forward from c50/c51**; no regression this cycle.
- **Rhythm section fix priority (operator UPDATE #3) DISCHARGED**: drums + bass now transcribed against real htdemucs stems with content metrics, not synthetic-mix placeholders. Transcription-truth situation materially improved even where low_band_corr flags overtone-rich bass (transcription is correct; rendering-fidelity gate needs c55 fluidsynth-GM-33 replacement).
- **Six-stem verdict path unblocked** contingent on peer clones-1+2 landing. RC10_LANDS aggregate is AND-gate over six per-stem accepts per rubric §D8.
- **c53 rc5 tempo estimates re-used cleanly** in D4 beat-grid snap postprocessing; cross-branch anchor reuse working as designed.
- **Content-metric gate discriminates over-transcription, not just under-transcription** (c53 clone-1 finding extended here): drums onset-F1 tautology at §D2(a) also demonstrates the same lemma from a different angle — reference-detector selection matters.
- **Fanout-scope discipline propagating**: clone-1 c53 close demonstrated the pattern; clone-0 c54 close now reinforces it. "One more cycle" trap avoided.
- **Honest-negative-finding discipline holding** at 10+ consecutive cycles.
- **Egress unchanged (17+ cycles)**: HTTP 429 + tv_embedded failure mode. c50 htdemucs_6s fetch OK anomaly remains isolated (not re-probed).

c29 state-machine lemma respected: peer sub-leaves under c50 v2 rubric chain; ledger topology stays a DAG.

c32 → c33 → c36 v2 → c39 v3 → c47 Branch B MIXED → c50 peer-supersede fanout-namespace + rubric-chain convention held: substantive M-RECREATE-2/* unsuffixed; infra families -clone-0 suffixed.

Collision-modeling arc: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Scope of this fanout clone fully discharged. [[BRANCH_COMPLETE]] emitted per no-null-cycle-validation role guidance; auditor decision **COMPLETE**.

[END OUTPUT]